Disease Detectives C - Disease Detectives - Monta Vista Science Olympiad (Div. C) - 11-14-2025
Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GEM)

Instructions (shown before students start the test)

Introduction (shown after students start the test)

1. (1.00 pts) Which of the following best describes a common source outbreak?

- A) Influenza spreading person-to-person
- B) Cholera from a contaminated water supply
- C) HIV transmitted sexually
- D) Measles in a school

2. (1.00 pts) The basic reproduction number (R0) measures:

- A) Mortality rate
- B) Incubation period
- C) Transmission potential of a disease
- D) Case fatality ratio

3. (1.00 pts) Which study design follows a group of individuals over time to observe outcomes?

- A) Case-control
- B) Cohort
- C) Cross-sectional
- D) Ecological

4. (1.00 pts) During an outbreak investigation, the first step is usually:

- A) Develop hypotheses
- B) Establish the existence of an outbreak
- C) Implement control measures
- D) Collect laboratory samples

5. (1.00 pts) Which measure represents the proportion of exposed individuals who develop disease?

- A) Attack rate
- B) Incidence rate
- C) Prevalence
- D) Mortality rate

6. (1.00 pts) A foodborne outbreak linked to a single meal is an example of:

- A) Continuous common source
- B) Propagated
- C) Sporadic
- D) Point source

7. (1.00 pts) Which of the following is a key feature of a propagated outbreak?

- A) Single peak in cases
- B) Multiple peaks over time
- C) Short incubation period
- D) Only one person affected

8. (1.00 pts) Which measure estimates the number of new cases in a population during a specific time?

- A) Incidence rate
- B) Prevalence
- C) Mortality rate
- D) Case-fatality ratio

9. (1.00 pts) In a case-control study, participants are selected based on:

- A) Exposure status
- B) Outcome status
- C) Random sampling
- D) Age

10. (1.00 pts) Which of the following is NOT an example of primary prevention?

- A) Vaccination
- B) Health education
- C) Quarantine of infected individuals
- D) Wearing masks during a pandemic

11. (1.00 pts) During which outbreak phase is the epidemic curve plotted?

- A) Before investigation
- B) During investigation
- C) After control measures
- D) Never

12. (1.00 pts) Which of the following describes herd immunity?

- A) All individuals are immune
- B) Immunity is temporary
- C) Enough people are immune to prevent disease spread
- D) Only children are immune

13. (1.00 pts) Which measure represents the proportion of deaths among diagnosed cases?

- A) Incidence rate
- B) Mortality rate
- C) Attack rate
- D) Case-fatality ratio

14. (1.00 pts) An epidemiologist wants to study the association between smoking and lung cancer. Which study design is most appropriate?

- A) Case-control study
- B) Cohort study
- C) Cross-sectional study
- D) Ecological study

15. (1.00 pts) Which type of study is retrospective in nature?

- A) Case-control
- B) Cohort
- C) Randomized controlled trial
- D) Cross-sectional

16. (1.00 pts) Which of the following is an example of an emerging infectious disease?

- A) Seasonal flu
- B) COVID-19
- C) Common cold
- D) Chickenpox

17. (1.00 pts) What is the incubation period?

- A) Time from onset to recovery
- B) Time from exposure to symptom onset
- C) Time from diagnosis to death
- D) Time from birth to immunity

18. (1.00 pts) Which is an example of a vector-borne disease?

- A) Cholera
- B) Influenza
- C) Malaria
- D) Tuberculosis

19. (1.00 pts) Which factor is considered a host factor in disease occurrence?

- A) Pathogen type
- B) Immunity status
- C) Environmental sanitation
- D) Water contamination

20. (1.00 pts) Which of the following is an environmental factor in disease transmission?

- A) Age
- B) Gender
- C) Genetic predisposition
- D) Temperature and humidity

21. (1.00 pts) Which term refers to the total number of cases in a population at a specific time?

- A) Incidence
- B) Attack rate
- C) Prevalence
- D) Mortality rate

22. (1.00 pts) Which of the following best describes a common source outbreak?

- A) Cholera from a contaminated water supply
- B) Influenza spreading person-to-person
- C) HIV transmitted sexually
- D) Measles in a school

(1 point each, 70 points total) Each term will be used only once

Active Immunity	Acute Exposure	Agent	Antigen	Secondary Attack Rate	Information Bias	Selection Bias
Crude Birth Rate	Index Case	Necessary Cause	Class Interval	Comparison Value	Confounding	Delayed Health Effect
Detection Limit	Dose-Response Relationship	Efficacy	Efficiency	Endemic Level	Epidemiologic Trial	Analytic Epidemiology

Applied Epidemiology	Descriptive Epidemiology	Exposure-Dose Reconstruction	Fomite	Health Promotion	Healthy Worker Effect	Host Factor
Active Immunity	Herd Immunity	Passive Immunity	Incidence	Incubation Period	In Vitro	In Vivo
Crude Mortality Rate	Infant Mortality Rate	Neonatal Mortality Rate	Postneonatal Mortality Rate	Normal Distribution	Odds Ratio	Outbreak
Common-Source Outbreak	Point-Source Outbreak	Propagated Outbreak	Pandemic	Prevalence	Reservoir	Risk Communication
Random Sample	Representative Sample	Standard Deviation	Analytic Study	Case-Control Study	Cohort Study	Experimental Study
Observational Study	Medical Surveillance	Passive Surveillance	Sentinel Surveillance	Synergistic Effect	Teratogen	Airborne Transmission
Vectorborne Transmissions	Vehicleborne Transmissions	Virulency	Contact tracing	Confidentiality	Healthy lifestyle	Measure of association

23. (1.00 pts) The systematic assessment of people exposed or potentially exposed to health hazards.

24. (1.00 pts) Manipulation of the association between an exposure and a health outcome by a third variable that is related to both.

25. (1.00 pts) The immunity that results from the production of antibodies by the immune system in response to the presence of an antigen.

26. (1.00 pts) The number of deaths of children from birth up to, but not including, 28 days per 1,000 live births.

27. (1.00 pts) Contact with a substance that occurs once or for only a short time.

28. (1.00 pts) A study that evaluates the association between exposure to hazardous substances and disease by testing scientific hypotheses.

29. (1.00 pts) Quantifies or assigns a numerical value to the strength of the statistical association between an exposure and outcome. Measures of association can be comparec

30. (1.00 pts)
Selected reporting units, with a high probability of seeing cases of the disease in question, good laboratory facilities and experienced well-qualified staff, identify and notify on certain diseases.

31. (1.00 pts)
The duty to keep patient information, such as medical records, secure. Confidentiality deals with how information will be used and who has access to this information (e.g., health care workers should not share confidential information with roommates, neighbors, or family members without the verbal or written consent of the patient).

32. (1.00 pts) A toxin or other foreign substance that induces an immune response in the body, especially the production of antibodies.

33. (1.00 pts)
The probability that infection occurs among susceptible persons within a reasonable incubation period following known contact with an infectious person or another infectious source.

34. (1.00 pts) Bias arising from measurement error.

35. (1.00 pts) The choice of individuals, groups or data for analysis in such a way that proper randomization is not achieved.

36. (1.00 pts) Gathers disease data from all potential reporting health care workers without prompting.

37. (1.00 pts)
The number of live births occurring among the population of a given geographical area during a given year, per 1,000 mid-year total population of the given geographical area durin
the same year.

38. (1.00 pts) A measure that is used to quantify the amount of variation of a set of data values.

39. (1.00 pts)
The initial patient in the population of an epidemiological investigation, or more generally, the first case of a condition or syndrome to be described in the medical literature.

40. (1.00 pts) A study that samples a group of people who share a defining characteristic, typically who experienced a common event in a selected period.

41. (1.00 pts) A factor that must be present for a disease to occur.

42. (1.00 pts)
A factor (e.g., a microorganism or chemical substance) or form of energy whose presence, excessive presence, or in the case of deficiency diseases, relative absence is essential f the occurrence of a disease or other adverse health outcome.

43. (1.00 pts) The size of each class into which a range of a variable is divided, as represented by the divisions of a histogram or bar chart.

44. (1.00 pts) Refers to the degree of damage caused by a microbe to its host.

45. (1.00 pts) Calculated concentration of a substance in air, water, food, or soil that is unlikely to cause harmful health effects in exposed people.

46. (1.00 pts)
An outbreak in which persons are exposed to the same source over a brief time, such as through a single meal or at an event. The number of cases rises rapidly to a peak and falls gradually.

47. (1.00 pts) A study that compares exposures of people who have a disease or condition with people who do not have the disease or condition.

48. (1.00 pts) A disease or an injury that happens as a result of exposures that might have occurred in the past.

49. (1.00 pts) The lowest concentration of a chemical that can reliably be distinguished from a zero concentration.

50. (1.00 pts) The relationship between the amount of exposure to a substance and the resulting changes in body function or health.

51. (1.00 pts) The ability of an intervention or program to produce the intended or expected results under ideal conditions.

52. (1.00 pts) The ability of an intervention or program to produce the intended or expected results with a minimum expenditure of time and resources.

53. (1.00 pts) The amount of a particular disease that is usually present in a community.

54. (1.00 pts) Consists of an external agent, a host and an environment in which host and agent are brought together, causing the disease to occur in the host.

55. (1.00 pts) The study of epidemiology that is concerned with the search for causes and effects, or the why and the how.

56. (1.00 pts)
The identification, monitoring, and support of a person who has been exposed to, and possibly infected with an infectious agent, such as a person who came in close contact with person with a confirmed or probable case of disease.

57. (1.00 pts)
A way of living that lowers the risk for disease and improves physical, mental, and social well-being, such as through healthy eating, regular physical activity, avoiding tobacco use, and reducing stress.

58. (1.00 pts) An outbreak which results from transmission from one person to another.

59. (1.00 pts) Putting epidemiological research methods to use in public health practice.

60. (1.00 pts) The study of the amount and distribution of a disease in a specified population by person, place, and time.

61. (1.00 pts) A method of estimating the amount of people's past exposure to hazardous substances.

62. (1.00 pts) Any nonliving object or substance capable of carrying infectious organisms, such as germs or parasites, and hence transferring them from one individual to anothe

63. (1.00 pts) A study where the researcher observes and systematically collects information, but does not try to change the people.

64. (1.00 pts) The process of enabling people to increase control over, and to improve, their health.

65. (1.00 pts)
A phenomenon where workers usually exhibit lower overall death rates than the general population because the severely ill and chronically disabled are ordinarily excluded from employment.

66. (1.00 pts) A biologic response to multiple substances where one substance worsens the effect of another substance.

67. (1.00 pts) Refers to the traits of an individual person or animal that affect susceptibility to disease, especially in comparison to other individuals.

68. (1.00 pts) The immunity that results from the production of antibodies by the immune system in response to the presence of an antigen.

69. (1.00 pts) A sample whose characteristics correspond to those of the original population or reference population.

70. (1.00 pts) The immunity to a pathogen in a population based on the acquired immunity to it by a high proportion of members over time.

71. (1.00 pts) A study where the researcher intervenes to change something (e.g., gives some patients a drug) and then observes what happens.

72. (1.00 pts) The short-term immunity that results from the introduction of antibodies from another person or animal.

73. (1.00 pts) The number of new cases of disease in a defined population over a specific time period.

74. (1.00 pts) The period between exposure and onset of clinical symptoms.

75. (1.00 pts) The exchange of information to increase understanding of health risks.

76. (1.00 pts) In an artificial environment outside a living organism or body.

77. (1.00 pts) Infections transmitted by the bite of infected arthropod species.

78. (1.00 pts) Within a living organism or body.

79. (1.00 pts) The mortality rate from all causes of death for a population during a specified time period.

80. (1.00 pts) The habitat in which an infectious agent normally lives, grows and multiplies

81. (1.00 pts) The number of deaths of infants under one year old per 1,000 live births.

82. (1.00 pts) A substance that causes defects in development between conception and birth.

83. (1.00 pts)
A set of items that have been drawn from a population in such a way that each time an item was selected, every item in the population had an equal opportunity to appear in the sample.

84. (1.00 pts) The number of deaths of children from 28 days up to, but not including, one year per 1,000 live births.

85. (1.00 pts) A function that represents the distribution as a symmetrical bell-shaped graph.

86. (1.00 pts) Infections transmitted by an inanimate object or material.

87. (1.00 pts) A measure of association between an exposure and an outcome.

88. (1.00 pts) A sudden increase in occurrences of a disease in a particular time and place.

89. (1.00 pts) An outbreak in which a group of persons are all exposed to an infectious agent or a toxin from the same source.

90. (1.00 pts) An epidemic of infectious disease that has spread through human populations across a large region.

91. (1.00 pts) When bacteria or viruses travel on dust particles or on small respiratory droplets that may become aerosolized when people sneeze, cough, laugh, or exhale.

92. (1.00 pts) The number of existing disease cases in a defined population during a specific time period.

93. (1.00 pts) Exposure to the infectious agent marks the beginning of what stage?

- A) Stage of susceptibility
- B) Stage of subclinical disease
- C) Stage of clinical disease
- D) Stage of disability

Match the diseases with their respective modes of transmission for the next 10 questions:

a. Airborne transmission

b. Water-borne transmission

c. Food-borne transmission

d. Vector transmission

94. (1.00 pts) Tuberculosis

(Mark **ALL** correct answers)

- A) a
- B) b,
- C) c
- D) d

95. (1.00 pts) Shigellosis

(Mark **ALL** correct answers)

- A) a
- B) b
- C) c
- D) d

96. (1.00 pts) SARS

(Mark **ALL** correct answers)

- A) a
- B) b
- C) c
- D) d

97. (1.00 pts) Chagas disease

(Mark **ALL** correct answers)

- A) a

- B) b
- C) c
- D) d

98. (1.00 pts) Giardiasis

(Mark **ALL** correct answers)
- A) a
- B) b
- C) c
- D) d

99. (1.00 pts) Hepatitis

(Mark **ALL** correct answers)
- A) a
- B) b
- C) c
- D) d

100. (1.00 pts) Lyme disease

(Mark **ALL** correct answers)
- A) a
- B) b
- C) c
- D) d

101. (1.00 pts) Varicella

(Mark **ALL** correct answers)
- A) a
- B) b
- C) c
- D) d

102. (1.00 pts) Typhoid fever

(Mark **ALL** correct answers)
- A) a
- B) b
- C) c
- D) d

103. (1.00 pts) Influenza

(Mark **ALL** correct answers)

- ☐ A) a
- ☐ B) b
- ☐ C) c
- ☐ D) d

104. (1.00 pts) Viruses, prions, and bacteria are examples of pathogens.

○ True ○ False

105. (1.00 pts) A reservoir for infectious disease always shows symptoms of the disease.

○ True ○ False

106. (1.00 pts) A respiratory infection is a result of indirect transmission.

○ True ○ False

107. (1.00 pts) An example of a vector is the deer tick that transmits Lyme disease.

○ True ○ False

108. (1.00 pts) The respiratory system can function as both a portal of entry and a portal of exit for disease transmission.

○ True ○ False

109. (1.00 pts) With most pathogens, people are not infectious during the incubation stage.

○ True ○ False

110. (1.00 pts) The clinical stage of infection is also called the acute stage.

○ True ○ False

111. (1.00 pts) A vaccination is an example of naturally acquired immunity.

○ True ○ False

112. (1.00 pts) Passively acquired immunity results when antibodies are introduced into the body.

○ True ○ False

113. (1.00 pts) There is conclusive evidence that the MMR immunization causes autism.

○ True ○ False

114. (1.00 pts) Heart Disease is still the number one killer in the world (not including Corona)

○ True ○ False

115. (1.00 pts) People age 65 and older survive an average of 14 to 18 years after a diagnosis of Alzheimer’s dementia

○ True ○ False

116. (1.00 pts) The flu virus, influenza, can live on surfaces for up to 5 hours

○ True ○ False

117. (1.00 pts) A flu shot can lower your risk of contracting acute bronchitis as well

○ True ○ False

118. (1.00 pts) Lung cancer is the biggest cancer killer in both men and women in the US.

○ True ○ False

119. (1.00 pts) Breast cancer is the most common cancer in American women

○ True ○ False

120. (1.00 pts) Cirrhosis is a disease that affects the liver

○ True ○ False

121. (1.00 pts) Males tend to have a higher risk of developing Alzheimer's Disease

○ True ○ False

122. (1.00 pts) In Parkinson's Disease, no two people have the same exact symptoms.

○ True ○ False

123. (1.00 pts) Chronic obstructive pulmonary disease (COPD) is mainly caused by pollutants in the workplace

○ True ○ False

124. (1.00 pts) Stroke is a leading cause of serious long-term disability.

○ True ○ False

125. (2.00 pts) A key feature of a cross-sectional study is that

(Mark **ALL** correct answers)

- ☐ A) It usually provides information on prevalence rather than incidence
- ☐ B) It is limited to health exposures and behaviors rather than health outcomes
- ☐ C) It is more useful for descriptive epidemiology than it is for analytic epidemiology
- ☐ D) It is synonymous with survey

126. (2.00 pts)
The Iowa Women's Health Study, in which researchers enrolled 41,837 women in 1986 and collected exposure and lifestyle information to assess the relationship between these factors and subsequent occurrence of cancer, is an example of which type(s) of study?

(Mark **ALL** correct answers)

- ☐ A) Experimental
- ☐ B) Observational
- ☐ C) Cohort
- ☐ D) Case-control
- ☐ E) Clinical trial

127. (3.00 pts)
Explain what impact it could have on public health, disease surveillance, and outbreak responses if hundreds of CDC Disease Detectives are laid off - especially considering how this might affect students and future training opportunities in epidemiology and public health careers.